KIANA GHASSABI

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ABOUT ME

PhD researcher in machine learning and multivariate time series, focused on modeling dynamic systems.
I enjoy working with real-time data, identifying patterns, and building intuition for how complex systems evolve. I am particularly interested in decision-making under uncertainty and how data-driven approaches can support fast and structured reasoning.

EDUCATION

Aarhus University
2024 - Present

PhD in Computer Engineering

Thesis topic: Transformer-Based Forecast-Guided Generative Modeling

- Developing machine learning models for multivariate time series forecasting and generation.
- Modeling complex, dynamic systems with a focus on stability and long-horizon behavior.
- Designing evaluation frameworks for real-world sequential data using statistical and spectral analysis to capture uncertainty and structural patterns.

IASBS, Zanjan, Iran
2020 - 2023

M.Sc. in Computer Science

Thesis topic: Improving the Efficiency of Secure Data Deduplication Methods in Cloud Storage Services

- Covered advanced topics in machine learning, data mining, cryptography.
- Studied the design and optimization of secure and scalable storage systems.
- Published research papers in journals and conferences.

IASBS, Zanjan, Iran
2016 - 2020

B.Sc. in Information Technology Engineering

Thesis topic: Improving the Decoding Speed of Packet Recovery in Network Coding

- Focused on foundational concepts in database management systems, operating systems, programming, and software engineering.
- Worked on projects involving system design, data structures, and algorithms.
- Explored real-world applications to optimize performance and reliability.

TEACHING EXPERIENCE

Aarhus university
|Aarhus, Denmark|

IASBS
|Zanjan, Iran|

Teaching Assistant

Responsibilities: Teaching, Holding Office Hours, Preparing Course Materials.

- **Communication Systems**| Bachelors level , 2025,2026
- **Distributed Storage** | Masters Level 2025
- **Advanced Computer Networks** | Masters Level 2022, 2023
- **Network Coding** | Masters Level 2020, 2021
- **Network Security** | Masters Level 2022
- **Computer Networks** | Bachelors level 2019, 2021

WORK EXPERIENCE

Wireless Communication Lab, IASBS
|Zanjan, Iran|
2020 – 2023

RESEARCHER

- Designed data-driven methods to analyze and optimize large-scale systems
- Applied statistical and algorithmic techniques to improve system reliability
- Extracted patterns from structured data to support decision-making
- Collaborated on research problems involving uncertainty and system dynamics

MetaverseAlliances
|Tehran, Iran|
2021 - 2022

Research and Development Team Lead

- Designed tokenomics models using quantitative reasoning and system-level thinking
- Analyzed market mechanisms and incentive structures in decentralized systems
- Developed data-driven strategies for ecosystem growth and sustainability
- Explored behavior of complex financial-like systems under changing conditions

IoT Lab, IASBS
|Zanjan, Iran|
2017- 2019

Embedded System Engineer

- Developed an LPWAN-based data monitoring system for HVAC, reducing energy usage by 60%.
- Handled end-to-end system programming using Raspberry Pi and Arduino.
- Demonstrated lab prototypes and software at industry exhibitions.

Sarve Saba
|Zanjan, Iran|
2016- 2017

Marketing Trainee

- Researched and analyzed market trends to identify emerging industry needs.
- Developed and implemented targeted marketing strategies to align with market demands.
- Monitored competitor activities to assess market positioning and identify opportunities for growth.

Beyondmegasoft
|London, England|
2019- 2020

RESEARCHER

- Designed and modeled token ecosystems with strong focus on user engagement metrics.
- Developed data-driven DeFi models (staking, lending, liquidity) aligned with market demand.

SKILLS

Languages

- Farsi (Native)
- Turkish (Native)
- English (Professional Proficiency)
- Danish (Beginner)

Programming Languages

- Python (Professional)
- C++ (Professional)
- R (Intermediate)
- Microcontroller Programming (Intermediate)
- MATLAB (Intermediate)
- Powershell (Beginner)

Data Science

- Data Visualization Tools
- Signal Processing
- Machine Learning Algorithms (Time series forecasting, supervised/unsupervised learning, and predictive modeling)

Technical

- LaTeX (Expert)
- IoT Sensors and Modules
- Version Controls (Git, GitHub)
- Qt Programming
- Microcontrollers
- Database Management
- Data Visualization
- Microsoft Office (Excel, Powerpoint)

Soft

- Analytical Thinking
- Flexibility
- Continuous Learning
- Self-Motivation
- Collaboration
- Conflict Resolution

PUBLICATIONS

1. K. Ghassabi, P. Pahlevani, "DEDUCT: A Secure Deduplication of Textual Data in Cloud Environments", in IEEE Access, May 2024, doi: 10.1109/ACCESS.2024.3402544.

2. K. Ghassabi, P. Pahlevani and D. Lucani, "Deduplication of Textual Data by NLP Approaches," at 2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring), Aug. 2023, doi: https://doi.org/10.1109/VTC2023-Spring57618.2023.10199538.

3. H. K. Nazari, **K. Ghassabi**, P. Pahlevani and D. Lucani, "Improving the Decoding Speed of Packet Recovery in Network Coding," in IEEE Communications Letters, vol. 25, no. 2, pp. 351-355, Feb. 2021, doi: 10.1109/LCOMM.2020.3031011.